

OSPFv3 for IPv6

Student Lab Workbook

Topics: Neighbor Adjacencies • Point-to-Point • Broadcast DR/BDR • Router ID

Purpose: Build a single OSPFv3 process for IPv6 and practice how OSPFv3 forms neighbor adjacencies on point-to-point and broadcast networks, how DR/BDR election works on a broadcast segment, and how the OSPFv3 Router ID is configured.

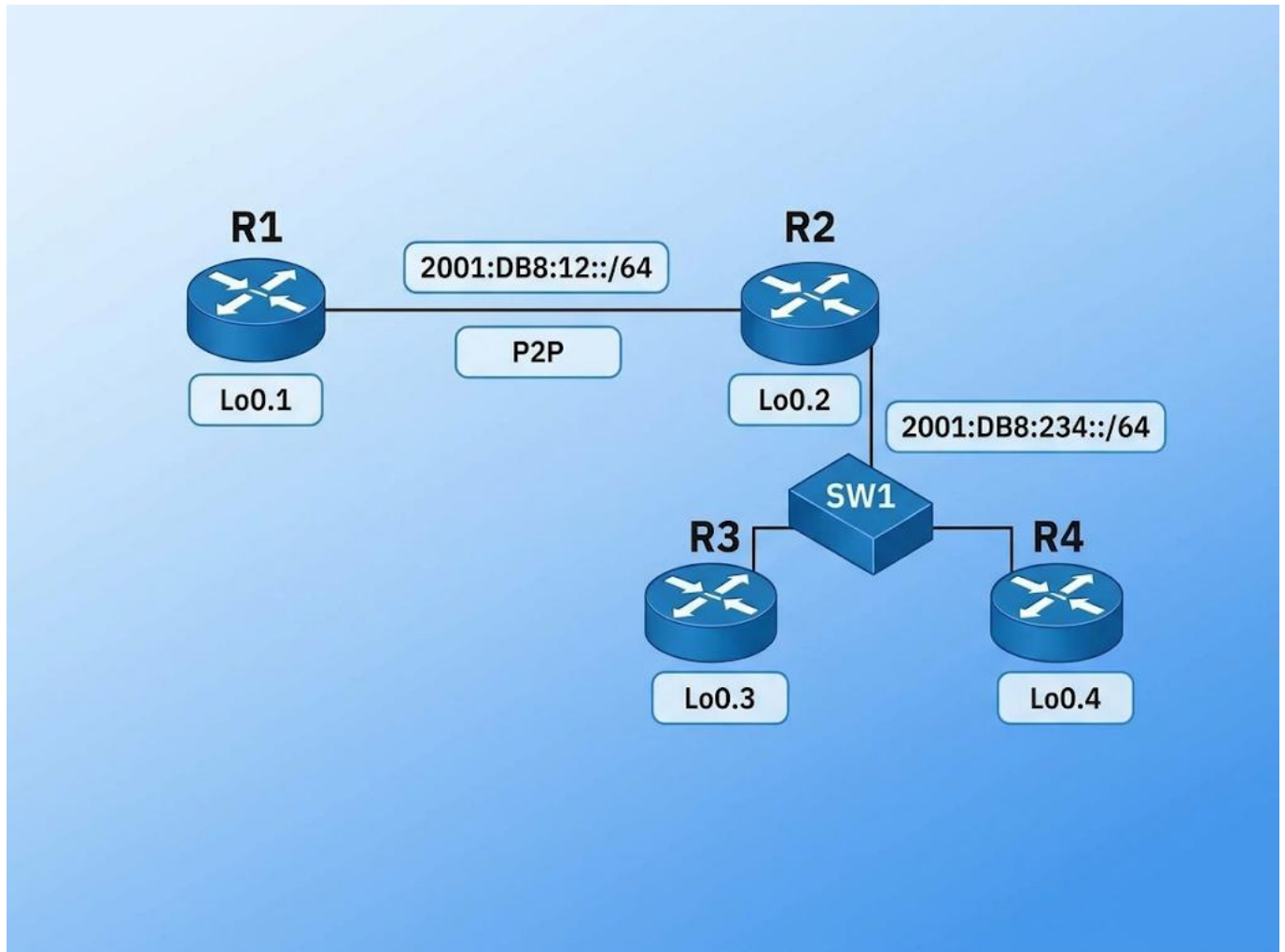
1. Learning Objectives

- Configure IPv6 addressing and enable IPv6 routing.
- Create one OSPFv3 process for IPv6 on all routers.
- Configure a unique OSPFv3 Router ID on every router.
- Form and verify OSPFv3 neighbor adjacencies.
- Understand OSPFv3 operation on a point-to-point network.

Lab Requirements

- Cisco Packet Tracer, Cisco CML, EVE-NG, or compatible Cisco IOS/IOS XE routers.
- Four routers: R1, R2, R3, and R4.
- IPv6-capable Ethernet interfaces.
- Console or SSH access to each router.
- One Layer-2 Ethernet broadcast segment for R2-R3-R4.

Topology



OSPFv3 Area: 0
Single OSPFv3 process: 10
R1-R2: point-to-point network
R2-R3-R4: broadcast Ethernet network
R2/R3/R4 participate in DR/BDR election

Physical implementation: Connect R1 directly to R2. Connect R2, R3, and R4 to the same switch/broadcast Ethernet segment. The diagram's lower connections represent the shared LAN, not direct router-to-router cables.

IPv6 Addressing Plan

Router	Interface	IPv6 Address	OSPF Area	Role
R1	G0/0	2001:DB8:12::1/64	0	Point-to-point
R2	G0/0	2001:DB8:12::2/64	0	Point-to-point
R2	G0/1	2001:DB8:234::2/64	0	Broadcast LAN
R3	G0/0	2001:DB8:234::3/64	0	Broadcast LAN
R4	G0/0	2001:DB8:234::4/64	0	Broadcast LAN

OSPFv3 Router-ID Plan

Router	Router ID	Expected Role on Broadcast LAN
R1	1.1.1.1	N/A
R2	2.2.2.2	Expected DR candidate
R3	3.3.3.3	Expected BDR candidate
R4	4.4.4.4	DROTHER candidate

OSPF router IDs are 32-bit values written in IPv4 dotted-decimal format. They are identifiers only; they do not have to be configured as IPv4 addresses on an interface.

```
Router(config)# interface GigabitEthernet0/1
```

```
Router(config-if)# ip address 192.168.10.2 255.255.255.0
```

```
Router(config-if)# ip ospf 1 area 0
```

```
router ospf 1
```

```
router-id 2.2.2.2
```

```
network 10.0.12.0 0.0.0.3 area 0
```

```
network 10.0.23.0 0.0.0.3 area 0
```

Basic IPv6 Connectivity

Configure IPv6 addressing first. Do not configure OSPFv3 yet.

R1

```
enable
configure terminal
hostname R1
ipv6 unicast-routing
interface g0/0
description P2P-to-R2
ipv6 address 2001:DB8:12::1/64
no shutdown
end
```

R2

```
enable
configure terminal
hostname R2
ipv6 unicast-routing
interface g0/0
description P2P-to-R1
ipv6 address 2001:DB8:12::2/64
no shutdown
interface g0/1
description Broadcast-LAN
ipv6 address 2001:DB8:234::2/64
no shutdown
end
```

R3

```
enable
configure terminal
hostname R3
ipv6 unicast-routing
interface g0/0
  description Broadcast-LAN
  ipv6 address 2001:DB8:234::3/64
  no shutdown
end
```

R4

```
enable
configure terminal
hostname R4
ipv6 unicast-routing
interface g0/0
  description Broadcast-LAN
  ipv6 address 2001:DB8:234::4/64
  no shutdown
end
```

Verify IPv6 Connectivity Before OSPFv3

```
show ipv6 interface brief
show ipv6 interface g0/0
ping 2001:DB8:12::2
ping 2001:DB8:234::2
ping 2001:DB8:234::3
ping 2001:DB8:234::4
```

Configure One OSPFv3 Process

Use one OSPFv3 process number, 10, on every router. Enable OSPFv3 directly under the participating IPv6 interfaces.

R1 — OSPFv3

```
configure terminal
ipv6 router ospf 10
  router-id 1.1.1.1
interface g0/0
  ipv6 ospf 10 area 0
end
```

R2 — OSPFv3

```
configure terminal
ipv6 router ospf 10
  router-id 2.2.2.2
interface g0/0
  ipv6 ospf 10 area 0
interface g0/1
  ipv6 ospf 10 area 0
end
```

R3 — OSPFv3

```
configure terminal
ipv6 router ospf 10
router-id 3.3.3.3
interface g0/0
ipv6 ospf 10 area 0
end
```

R4 — OSPFv3

```
configure terminal
ipv6 router ospf 10
router-id 4.4.4.4
interface g0/0
ipv6 ospf 10 area 0
end
```

Note: On some IOS/IOS XE releases the exact show-command spelling can vary. If 'show ospfv3 ...' is unavailable, use the OSPFv3 commands offered by your platform's CLI.

Neighbor Adjacencies

Wait for OSPFv3 to converge, then verify neighbors. The expected neighbor relationships are R1-R2 and the three-router broadcast segment involving R2, R3, and R4.

```
show ospfv3 neighbor
show ospfv3 interface brief
show ospfv3 interface g0/0
show ipv6 route ospf
```

Expected Observations

Link / Segment	Expected OSPFv3 Behavior
R1-R2	Neighbor adjacency forms over the point-to-point link.
R2-R3-R4 shared Ethernet	Neighbors form on the broadcast network.
Broadcast LAN	A DR and BDR are elected.
OSPFv3 process	All routers use process 10 and Area 0.

Point-to-Point Network

The R1-R2 link is treated as a point-to-point OSPF network. There are only two OSPF routers on this segment, so there is no DR/BDR election for this link.

```
R1# show ospfv3 interface g0/0
R2# show ospfv3 interface g0/0
R1# show ospfv3 neighbor
R2# show ospfv3 neighbor
```

Observe the network type reported for the interface. On platforms where the interface defaults to point-to-point for the chosen media, the adjacency forms without DR/BDR election.

```
ipv6 ospf network point-to-point
```

Broadcast Network and DR/BDR Election

```
R2# show ospfv3 interface g0/1
R3# show ospfv3 interface g0/0
R4# show ospfv3 interface g0/0
R2# show ospfv3 neighbor
```

Expected DR/BDR Result

Router	Router ID	Expected Broadcast Role
R2	2.2.2.2	DR
R3	3.3.3.3	BDR
R4	4.4.4.4	DROTHER

Correction: OSPF DR election compares interface priority first, then Router ID. Therefore, if all priorities are equal, the highest Router ID wins DR and the second-highest wins BDR. With the IDs listed above, the actual expected order is R3 = DR, R4 = BDR, R2 = DROTHER.

Router	Router ID	Expected Role
R1	1.1.1.1	Point-to-point only
R2	2.2.2.2	DROTHER
R3	3.3.3.3	DR
R4	4.4.4.4	BDR

This gives students a clear demonstration of Router ID selection without mixing OSPFv2 commands into the OSPFv3 lab.

Expected result on the R2-R3-R4 broadcast LAN: R3 = DR, R4 = BDR, R2 = DROTHER.

```
Final Router-ID plan:
R1 = 1.1.1.1
R2 = 2.2.2.2
R3 = 3.3.3.3
R4 = 4.4.4.4
```

Do not configure IPv4 OSPF priority commands in this OSPFv3-only lab. Keep the default/equal interface priority and use the Router IDs to demonstrate the election process. The highest eligible Router ID becomes DR and the second-highest becomes BDR.

Router ID Practice

Router ID is especially important in OSPFv3 because OSPFv3 can operate without IPv4 addresses on the participating interfaces, but the OSPF process still needs a 32-bit Router ID.

```
show ospfv3
show ospfv3 neighbor
```

Verify Learned IPv6 Routes

```
show ipv6 route
show ipv6 route ospf
show ospfv3 database
show ospfv3 neighbor
show ospfv3 interface brief
```

End-to-End Connectivity Test

Use loopback interfaces to create additional IPv6 prefixes and verify that OSPFv3 advertises them. Configure one loopback on each router:

```
R1:
interface loopback0
  ipv6 address 2001:DB8:1::1/128
  ipv6 ospf 10 area 0
```

```
R2:
interface loopback0
  ipv6 address 2001:DB8:2::2/128
  ipv6 ospf 10 area 0
```

```
R3:
interface loopback0
  ipv6 address 2001:DB8:3::3/128
  ipv6 ospf 10 area 0
```

```
R4:
interface loopback0
  ipv6 address 2001:DB8:4::4/128
  ipv6 ospf 10 area 0
```

Test:

```
R1# ping 2001:DB8:4::4
R4# ping 2001:DB8:1::1
R2# show ipv6 route ospf
R3# show ipv6 route ospf
```